A = 0101
B = 0110
Op = 11

COMPUTER ORGANIZATION · DIGITAL LOGIC

Designing a 4-bit ALU

AND, OR and Set-Less-Than (SLT), selected by a multiplexer — design, working and applications

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What the ALU does

The Arithmetic Logic Unit is the combinational heart of a CPU: it takes two 4-bit operands, A and B, computes several results in parallel, and a **multiplexer** picks the one requested by the operation-select lines.

&

AND

Bitwise: each result bit is 1 only when both input bits are 1.

```
0101 AND 0110 = 0100
```

|

OR

Bitwise: each result bit is 1 when either input bit is 1.

```
0101 OR 0110 = 0111
```

<

SLT

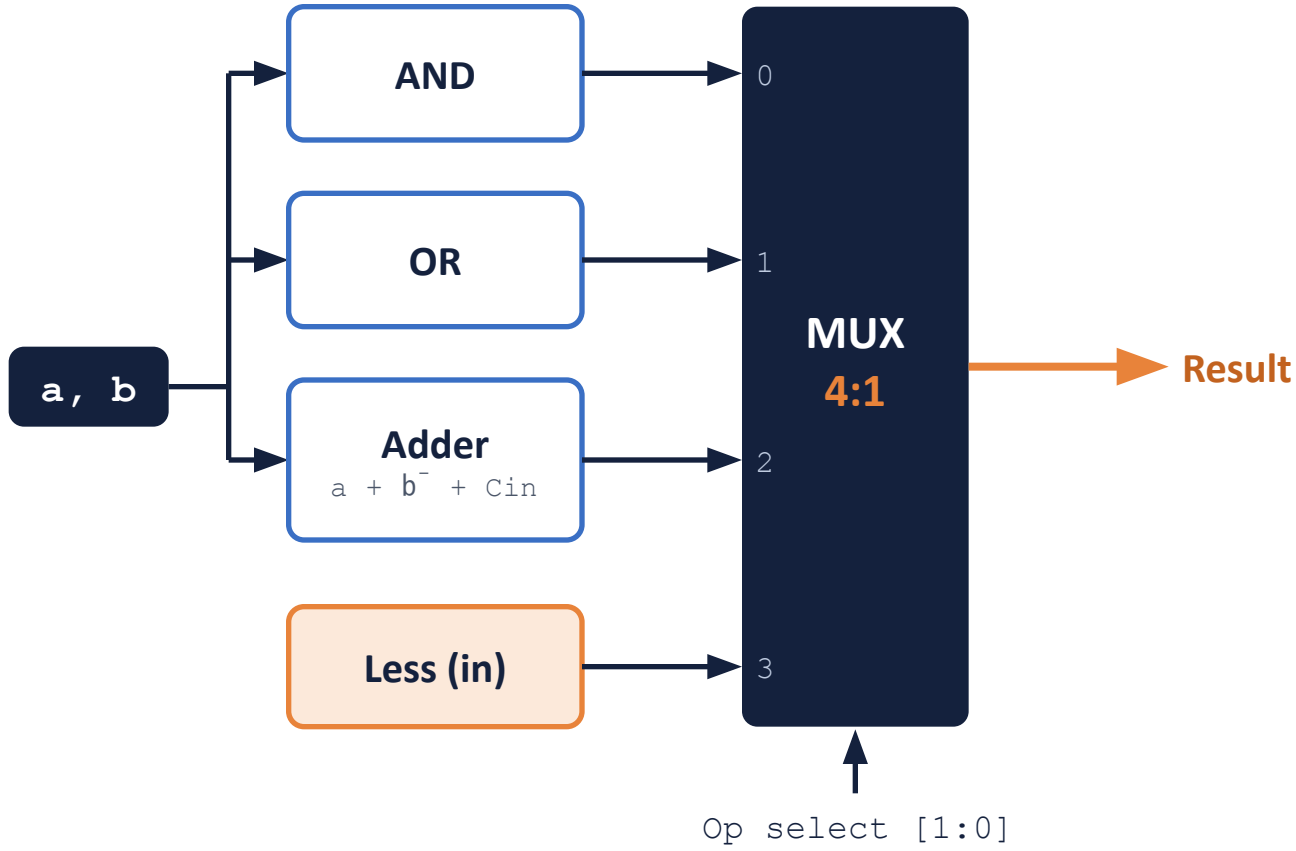
Set-Less-Than: result is 0001 if A < B (signed), otherwise 0000.

```
5 < 6  →  0001
```

The 1-bit ALU slice

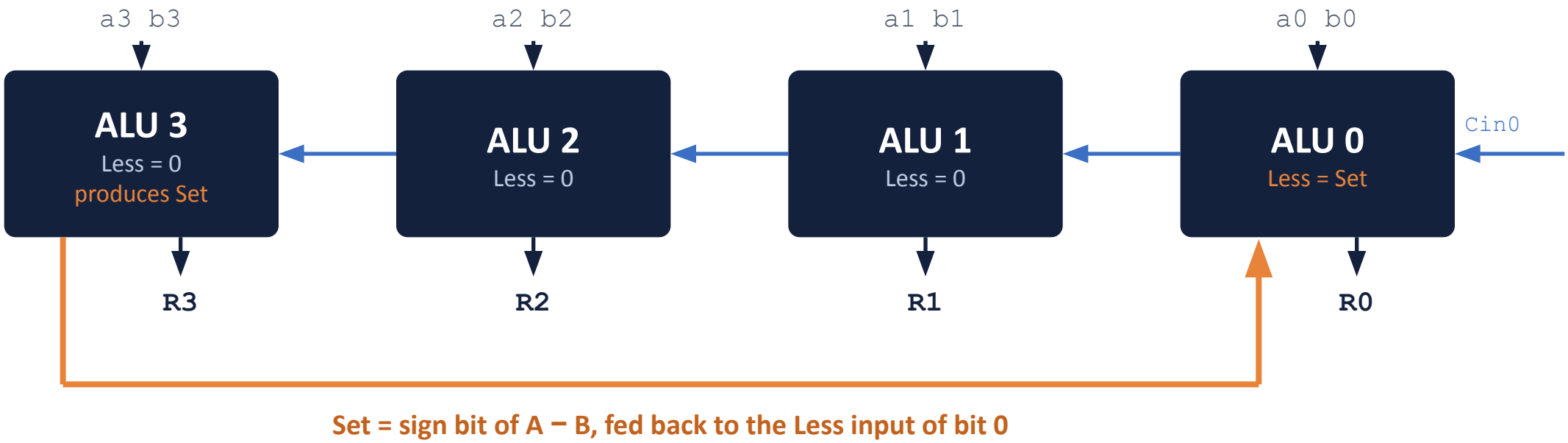
Each bit position has its own slice. An AND gate, an OR gate and a full adder all work on inputs a and b at once.

Binvert feeds $\bar{b}$ to the adder so it computes a − b for SLT.
A **4-to-1 MUX** with a 2-bit select chooses which signal becomes the slice's result.



Chaining four slices into a 4-bit ALU

Carries ripple from bit 0 to bit 3, and Op select and Binvert reach every slice. For SLT, CarryIn0 = 1 so the adders compute A − B.



How the select lines drive each operation

Operation	Binvert	Op[1:0]	Output
AND	0	00	A AND B
OR	0	01	A OR B
SLT	1	11	0001 if A < B, else 0000

For SLT, Binvert = 1 and CarryIn0 = 1 make every adder compute A + B + 1 = A − B. The MUX in each slice passes its Less input, so only bit 0 can be 1.

Worked example

```
A = 0101 (5)
B = 0110 (6)

AND → 0100
OR  → 0111

SLT: 0101 + 1001 + 1
   = 1111 → sign 1

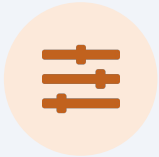
SLT → 0001 (5 < 6)
```

Where this ALU is used



CPU instructions

Implements MIPS/RISC-V and, or and slt; slt plus a branch gives if (a < b) decisions.



Bit masking

AND clears or tests bits; OR sets flags in control and status registers.



Comparison & sorting

SLT finds min/max values and drives sorting in embedded and DSP chips.



Scalable building block

The same slice repeats to build 8-, 16- or 32-bit ALUs in FPGAs and processors.

Thank You